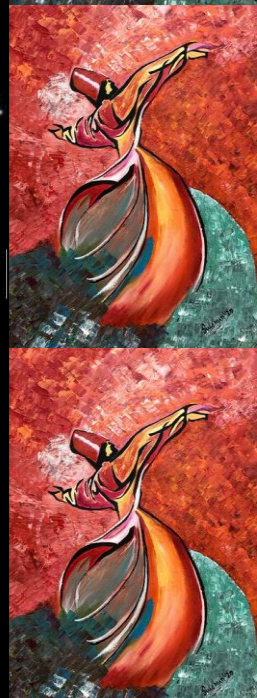




EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Spring 2026



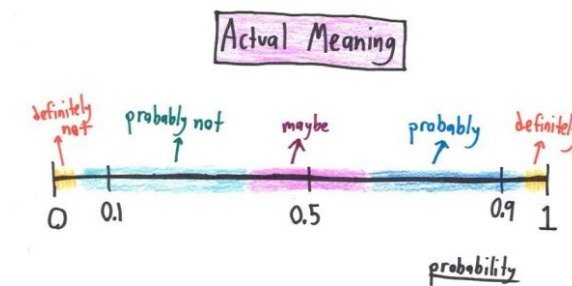


EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

Aamir Hasan

Week 10 - Lecture No 20 – March 18, 2026



Announcements!

1. Office hours (post Ramadan) – **Tuesday & Thursday 12:00 to 1:00 PM**
2. Quiz No 06 – April 01, 2026
3. Online class on March 25, 2026 (accommodation is requested)
 1. Cameras to remain switched on during the entire lecture
 2. Please ensure audio and video is working (see below)
 3. I will place questions to the students and not responding w/ audio & video would be used for participation & attendance. In cases where I am unable to ask questions to individuals, attendance would be taken sometime in between the lecture.
 4. Slides for the online class will be posted (after some delay)

4. TA's EHSAS Hours

EHSAS TUTORS' RAMADAN SCHEDULE – Spring 2026							
Course	Tutor	Email	Monday	Tuesday	Wednesday	Thursday	Friday
Intro to Probability & Statistics	Rayyan Ali Shah	rs08770@st.habib.edu.pk	2:00-3:00		2:00-3:00		12:40-1:40
	Hiba Aiman Ali	ha09269@st.habib.edu.pk	12:45-2:30		12:45-2:30		11:00-12:00
	Zain Naqi	zn09224@st.habib.edu.pk		9:00 - 10:30	11:45 - 12:45	9:00 - 10:30	

Agenda for today

1. Quiz No 05
2. Unit 5 – Continuous Random Variable

An apology!

"Before we start the quiz, I want to address something from our last session. I realized that I let my frustration get the better of me. That was not fair to you, and I apologize for creating a tense environment. I am passionate about this subject, but I need to manage that better. Let's have a productive class today."



Eid Ul Fitr

Messages, Wishes
to Students

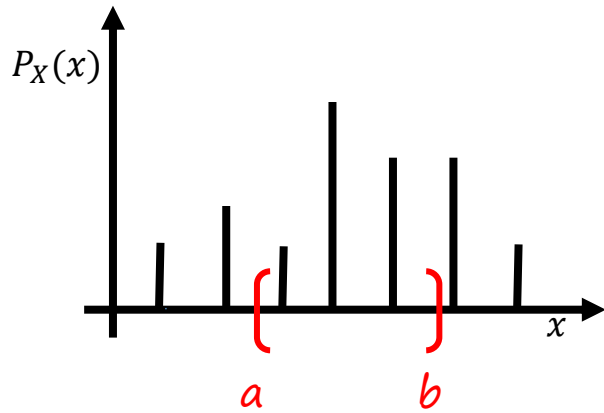


Quiz No 05

20:00

Unit 5
Continuous Random Variables

Discrete RVs

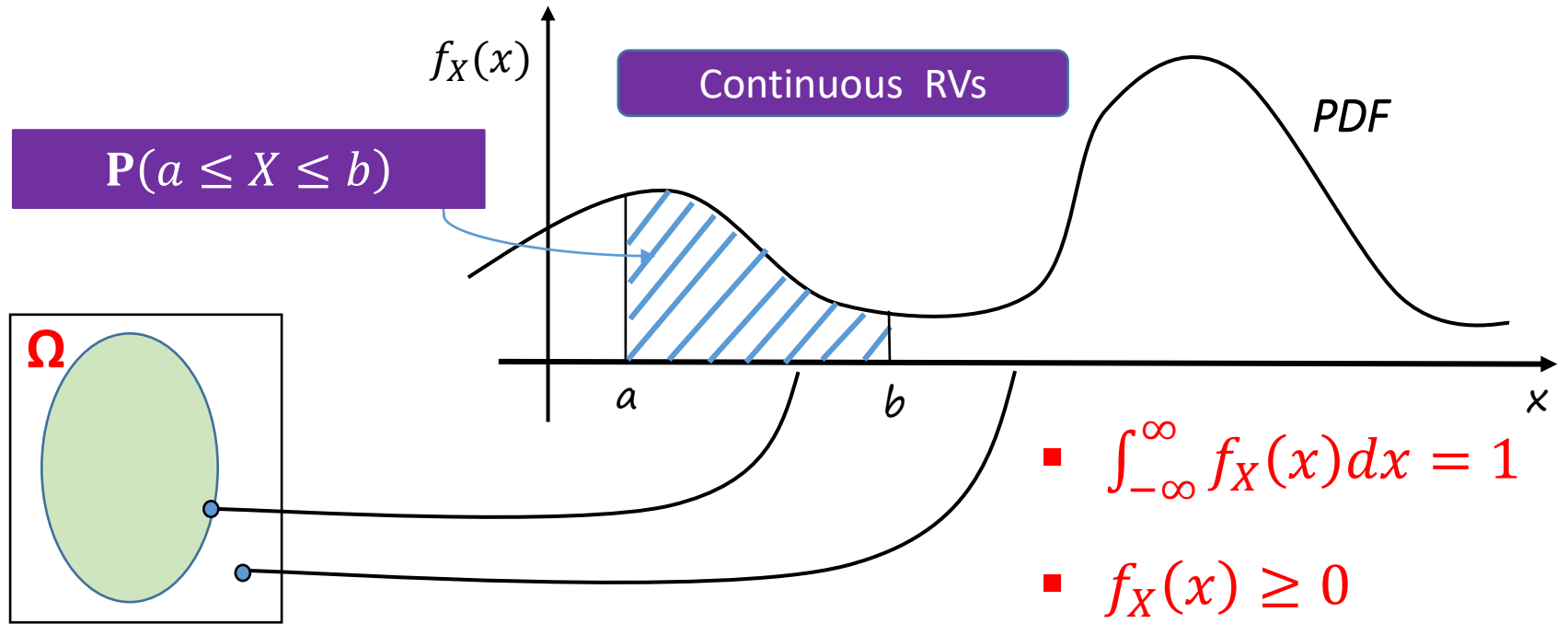


- $\sum_x p_X(x) = 1$
- $p_X(x) \geq 0$

$$P(a \leq X \leq b) = \sum_{x: a \leq x \leq b} p_X(x)$$

Probability Density Functions (PDFs)

Continuous RVs



- $\int_{-\infty}^{\infty} f_X(x) dx = 1$
- $f_X(x) \geq 0$

Definition: A random variable is **continuous** if it can be described by a PDF

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx$$

$$P(X = a) = 0$$

- *Aside - $P(1 \leq X \leq 3 \text{ or } 4 \leq X \leq 5) = P(1 \leq X \leq 3) + P(4 \leq X \leq 5)$*

Probability Density Functions (PDFs):

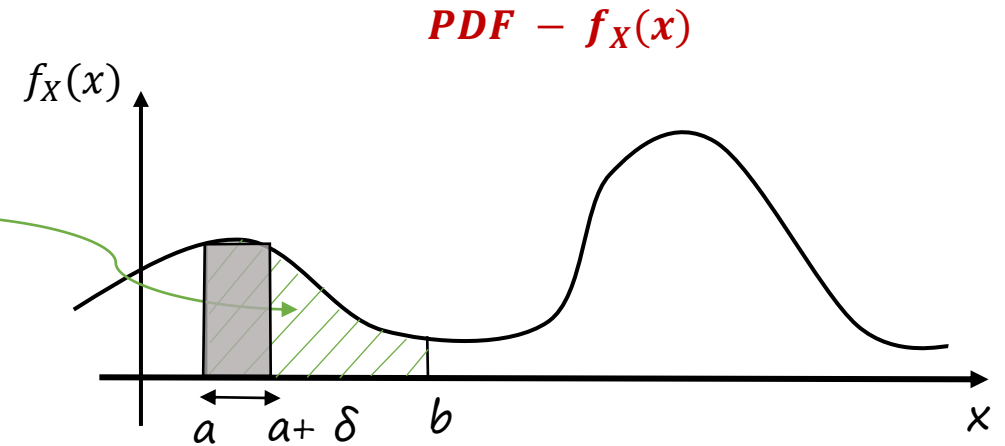
$f_X(a)$??

$\delta > 0$, small (approaching zero)



$$\mathbf{P}(a \leq X \leq a + \delta) \approx f_X(a)\delta$$

$\mathbf{P}(a \leq X \leq b)$



$$\blacksquare \int_{-\infty}^{\infty} f_X(x) dx = 1$$

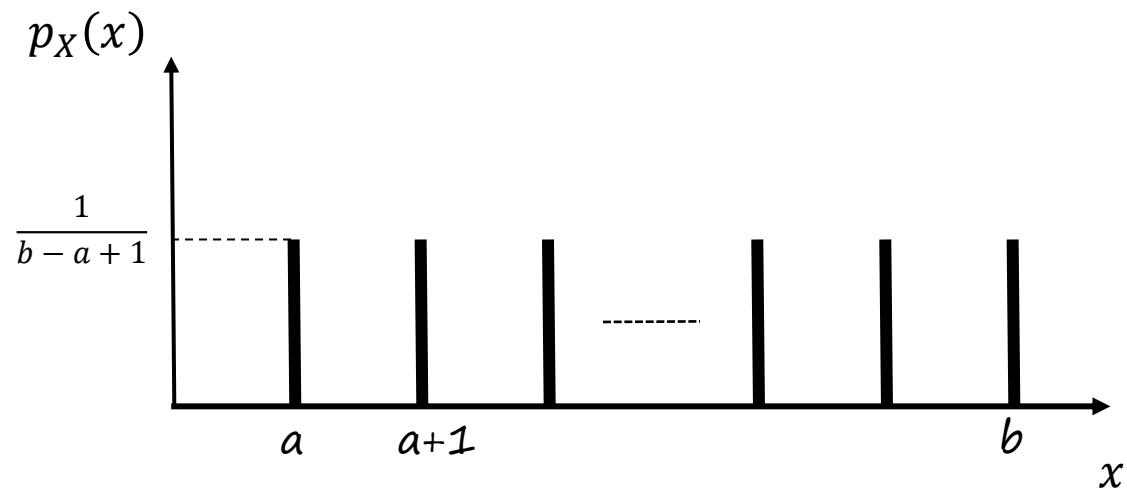
$$\blacksquare f_X(x) \geq 0$$

$$\mathbf{P}(a \leq X \leq b) = \int_a^b f_X(x) dx$$

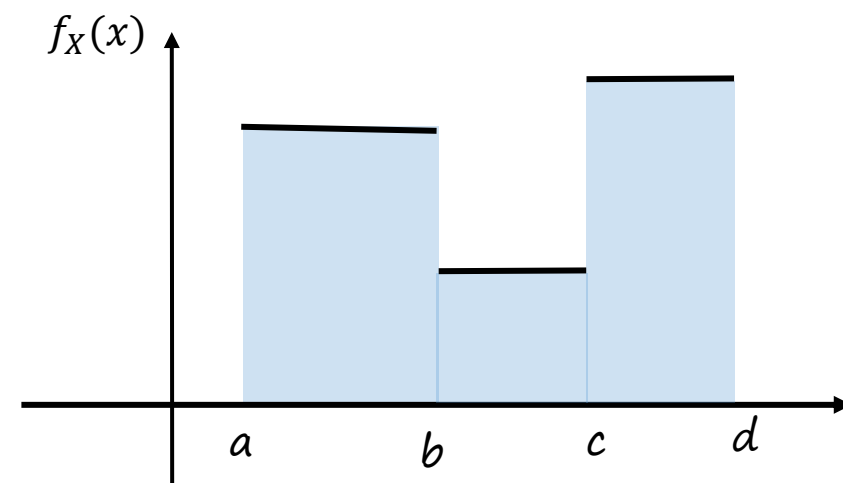
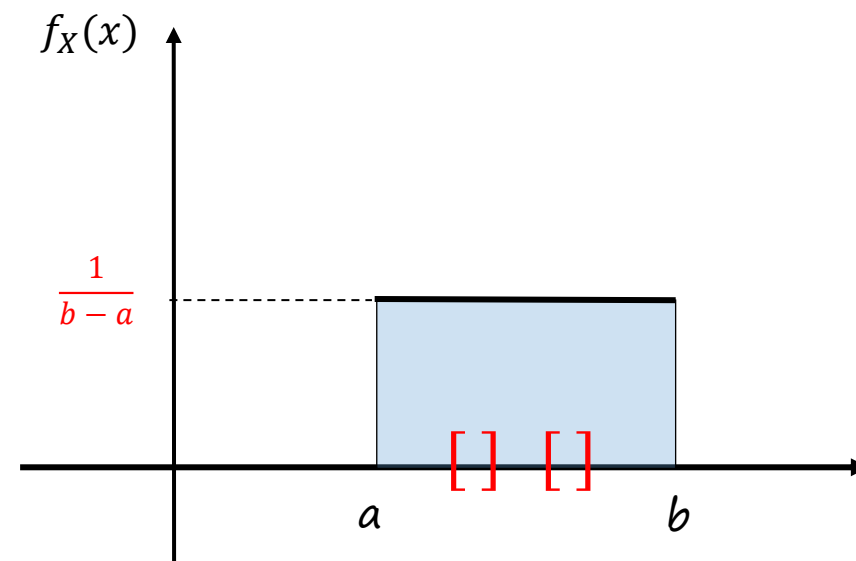
 $\mathbf{P}(a \leq X \leq b) = \mathbf{P}(\cancel{x=a}) + \mathbf{P}(\cancel{x=b}) + \mathbf{P}(a < X < b)$

$$\mathbf{P}(a \leq X \leq b) = \mathbf{P}(a < X < b)$$

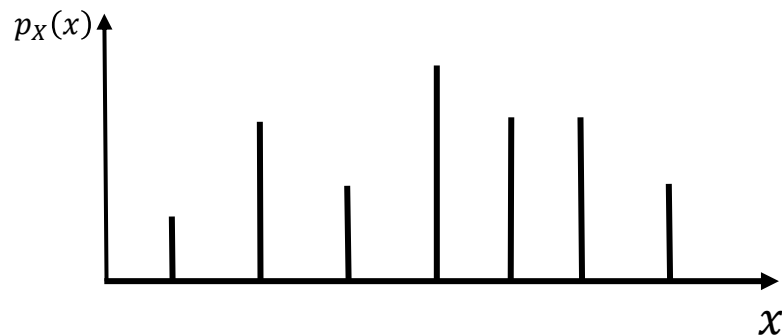
Example: continuous uniform PDF



- Generalization: piecewise constant PDF

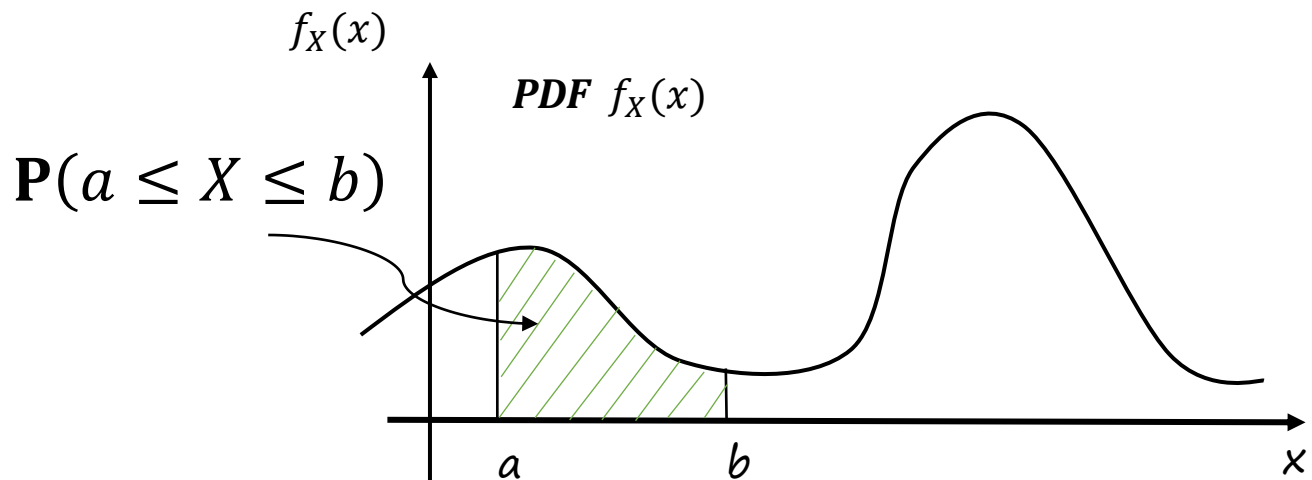


Expectation of a continuous RV



→ $E[X] = \sum_x x p_X(x)$

- **Interpretation:** Average in large number of independent repetitions of the experiment



→ $E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$

Assume, $\int_{-\infty}^{\infty} |x| f_X(x) dx < \infty$

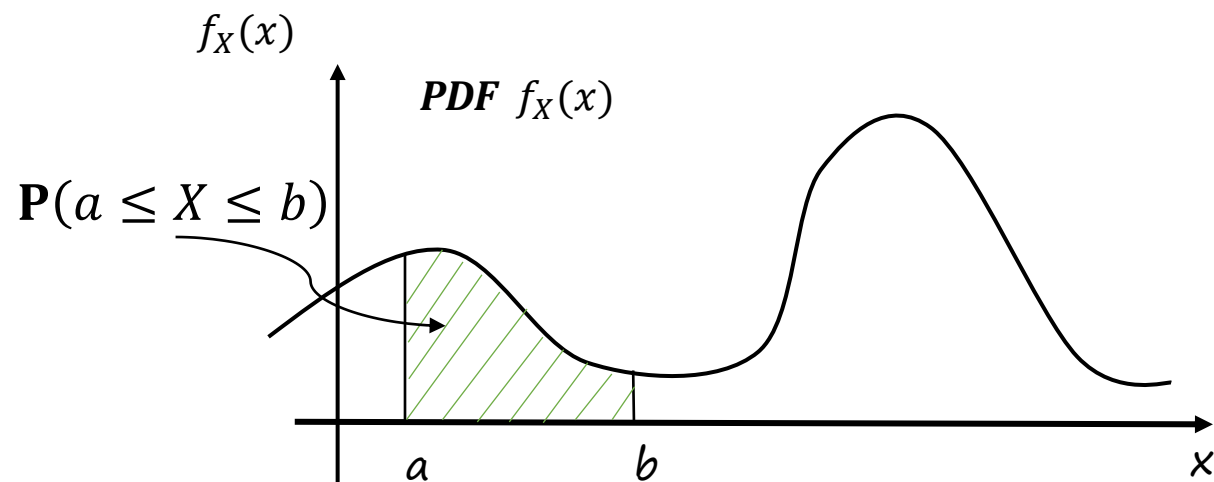
Expectation of a continuous RV

Assume, $\int_{-\infty}^{\infty} |x| f_X(x) dx < \infty$?

A continuous RV with an infinite mean is one for which the expected value, calculated by an integral (A), diverges. *The most common example is the Pareto distribution, where PDF has a "long tail" that extends indefinitely, causing the weighted average to become infinitely large*

Example: Pareto distribution

- The Pareto distribution is a classic example used in fields like economics to model phenomena like wealth or income.
- Its probability density function (PDF) is $f(x) = \frac{\alpha}{x^{\alpha+1}}$ for $x \geq 1$ and $\alpha > 0$.
- The mean of this distribution is the expected value, calculated by the integral $\int_1^{\infty} x \cdot f(x) dx$. (A)
- This integral only converges to a finite value if $\alpha > 1$. When $\alpha \leq 1$, the integral diverges to infinity. This means the average value of a Pareto-distributed variable is infinite if the parameter α is 1 or less.



$$\rightarrow E[X] = \int_{-\infty}^{\infty} x f_X(x) dx \quad \text{(A)}$$